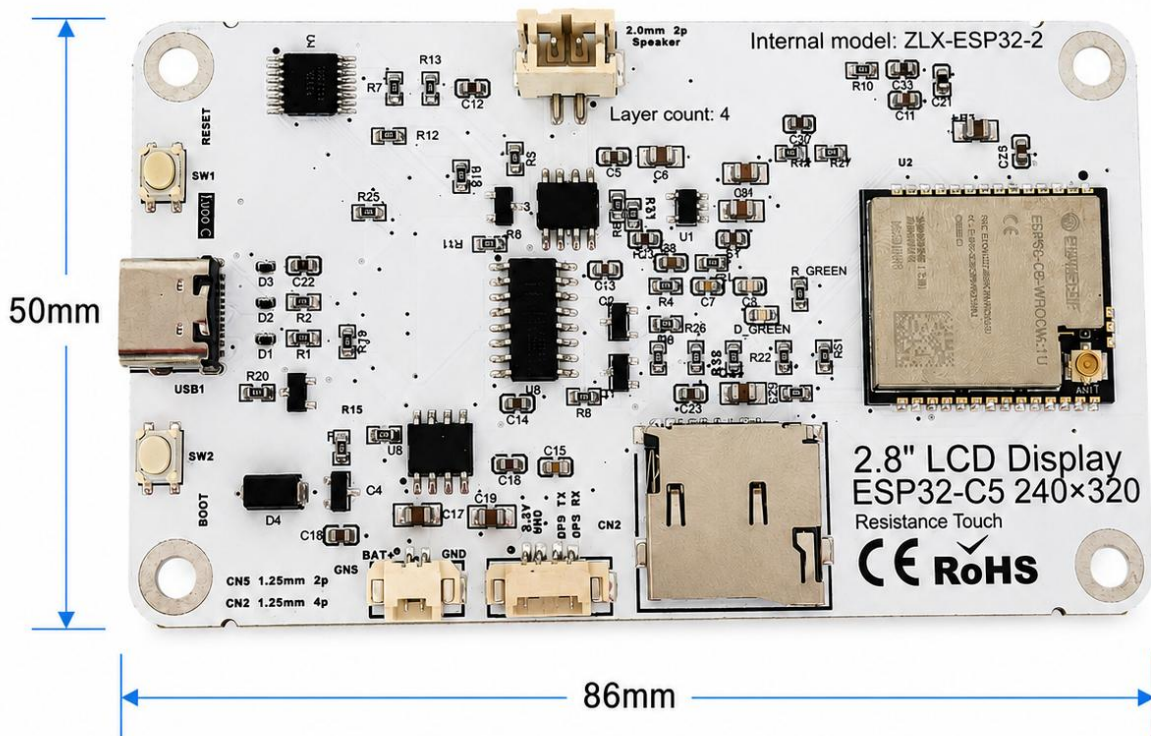


ZLX-ESP32-2

ESP32-C5 2.8-inch Touch Display Development Board

中英文用户说明书 / Bilingual User Manual



Model / 型号: ZLX-ESP32-2

Module / 模组: ESP32-C5-WROOM-1U-N8R8

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1. 产品概述 / 1. Product Overview

ZLX-ESP32-2 是一款基于 ESP32-C5-WROOM-1U-N8R8 模组的 2.8 英寸电阻式触摸屏开发板，集成彩色 LCD、XPT2046 触摸、microSD/TF 卡槽（仅提供卡槽，不附带 microSD/TF 卡）、GPS 接口、音频输出、电池电压检测、绿色 LED、BOOT/RESET 按键及外置天线接口。适合 Wi-Fi 6、Bluetooth LE、Zigbee/Thread、人机界面、数据记录及物联网应用开发。

ZLX-ESP32-2 is a 2.8-inch resistive-touch development board based on the ESP32-C5-WROOM-1U-N8R8 module. It integrates a color LCD, XPT2046 touch controller, a microSD/TF card slot (slot only; no microSD/TF card is included), GPS interface, audio output, battery-voltage sensing, green LED, BOOT/RESET buttons, and an external-antenna connector. It is suitable for Wi-Fi 6, Bluetooth LE, Zigbee/Thread, HMI, data logging, and IoT development.

主要特点 / Key Features

- ESP32-C5 32 位 RISC-V 单核处理器，最高 240 MHz / ESP32-C5 32-bit RISC-V single-core CPU, up to 240 MHz
- ESP32-C5-WROOM-1U-N8R8: 8 MB Flash + 8 MB PSRAM（模组板载资源，不是 microSD/TF 卡） / ESP32-C5-WROOM-1U-N8R8: 8 MB Flash + 8 MB PSRAM (onboard module memory, not a microSD/TF card)
- 2.4 GHz + 5 GHz 双频 Wi-Fi 6 (802.11ax) / 2.4 GHz + 5 GHz dual-band Wi-Fi 6 (802.11ax)
- Bluetooth LE、IEEE 802.15.4、Zigbee 3.0、Thread 1.4 / Bluetooth LE, IEEE 802.15.4, Zigbee 3.0, Thread 1.4
- 2.8 英寸 240×320 LCD，ST7789 驱动，BGR 初始化，不开启反色 / 2.8-inch 240×320 LCD, ST7789 driver, BGR initialization, display inversion disabled
- XPT2046 电阻式触摸控制器 / XPT2046 resistive touch controller
- LCD 与 TF 卡共用 SPI 总线，独立 CS；触摸使用独立软件 SPI / LCD and TF card share SPI with separate CS; touch uses an independent software SPI
- 板卡尺寸约 86 mm × 50 mm / Board size approx. 86 mm × 50 mm
- IPEX1 外置天线接口；使用无线功能前需正确连接外置天线 / IPEX1 external antenna connector; connect a suitable external antenna before radio use

适用对象 / Intended Users

适用于嵌入式开发者、物联网开发者、创客、教学实验及产品原型设计。

For embedded developers, IoT developers, makers, education/lab use, and product prototyping.

2. 外观与尺寸 / 2. Appearance and Dimensions

板卡外形尺寸约为 86 mm × 50 mm。以下正反面图片用于识别屏幕、接口与主要器件。

The board outline is approximately 86 mm × 50 mm. The images below help identify the display, connectors, and major components.

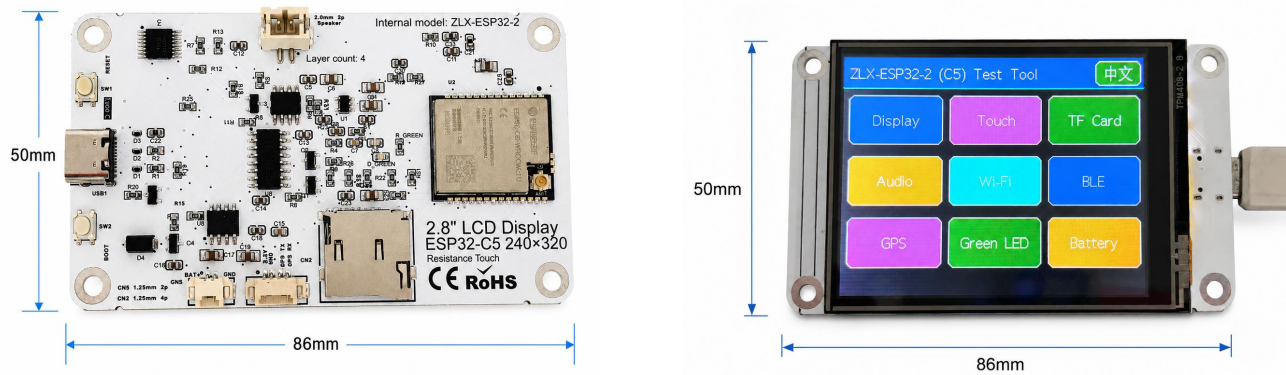


图 1 正面与屏幕面尺寸示意 / Figure 1 Board front and display-side dimensions

接口标注 / Connector Identification

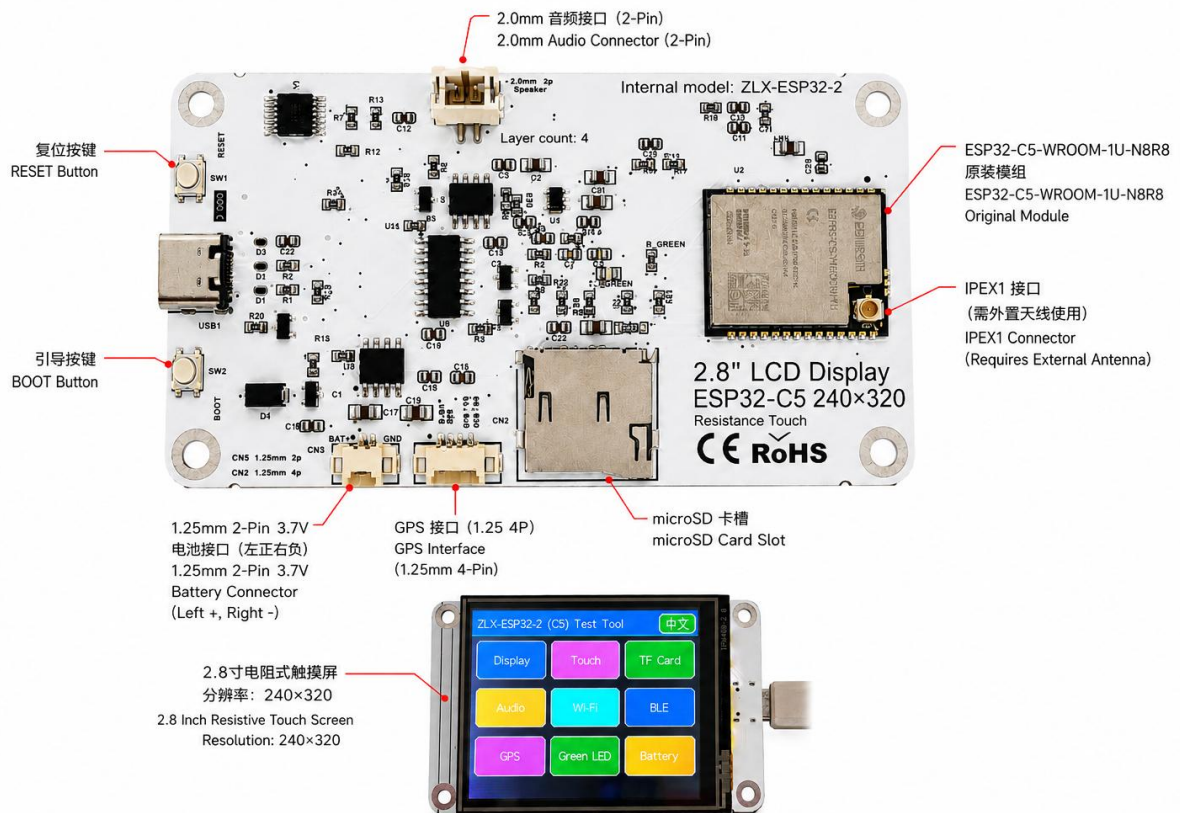


图 2 ZLX-ESP32-2 中英文接口标注 / Figure 2 Bilingual connector identification

3. 开发板规格 / 3. Board Specifications

项目 / Item	规格 / Specification
开发板型号 / Board model	ZLX-ESP32-2
核心模组 / Core module	ESP32-C5-WROOM-1U-N8R8
板卡尺寸 / Board size	Approx. 86 × 50 mm
显示屏 / Display	2.8-inch TFT LCD, ST7789, 240×320 native resolution
颜色顺序 / Color order	BGR initialization
显示反色 / Display inversion	OFF / Normal display mode（不开反色）
触摸 / Touch	Resistive touch / 电阻式触摸, XPT2046
microSD/TF 卡槽 / microSD/TF Card Slot	SPI mode; card not included / SPI 模式；不附带卡
GPS 接口 / GPS interface	1.25 mm 4-pin connector; UART signals on GPIO0/GPIO9
音频接口 / Audio	2.0 mm 2-pin speaker/audio connector; PWM audio + amplifier enable
电池接口 / Battery	3.7 V 1S, 1.25 mm 2-pin; protected 1000–3000 mAh recommended / 单节 3.7 V；建议 1000–3000 mAh 带保护板
外置天线 / External antenna	IPEX1 connector; required for the WROOM-1U radio path
按键 / Buttons	RESET, BOOT
状态指示 / Indicator	Green LED on GPIO26

说明 / Note: 屏幕原生分辨率为 240×320；当软件设置为横屏旋转时，逻辑显示尺寸可表现为 320×240。 / The panel native resolution is 240×320; in landscape rotation, the logical display size may be 320×240.

4. ESP32-C5-WROOM-1U-N8R8 模组参数 / 4. ESP32-C5-WROOM-1U-N8R8 Module Specifications

以下关键参数依据 Espressif 官方 ESP32-C5-WROOM-1 / WROOM-1U Datasheet v1.2。
The following key parameters are based on Espressif's official ESP32-C5-WROOM-1 / WROOM-1U Datasheet v1.2.

项目 / Item	规格 / Specification
内置芯片 / Embedded SoC	ESP32-C5HR8
CPU	32-bit RISC-V single core, up to 240 MHz
ROM	320 KB
HP SRAM	384 KB
LP SRAM	16 KB
Flash	8 MB Quad SPI
PSRAM	8 MB Quad SPI

项目 / Item	规格 / Specification
Wi-Fi	1T1R, 2.4 GHz + 5 GHz dual-band; IEEE 802.11ax/ac/b/g/n; up to 150 Mbps
2.4 GHz 频率 / Frequency	2412–2484 MHz
5 GHz 频率 / Frequency	5180–5885 MHz (subject to regional regulations)
Bluetooth	Bluetooth LE; Espressif v1.2 feature list states Bluetooth Core 6.0 certified
IEEE 802.15.4	2.4 GHz OQPSK, 250 Kbps
Zigbee / Thread	Zigbee 3.0 / Thread 1.4
GPIO	Up to 22 GPIOs at module level
典型外设 / Peripherals	GPIO, SPI, UART, I2C, I2S, RMT, LED PWM, USB Serial/JTAG, MCPWM, GDMA, CAN FD, SDIO slave, ADC, temperature sensor, timers, watchdogs, etc.
模组工作电压 / Module supply	3.0–3.6 V, typical 3.3 V
推荐电源能力 / Recommended supply capability	External supply should be capable of at least 0.6 A for the module
环境温度 / Ambient temperature	-40 to +85 °C
模组尺寸 / Module size	18.0 × 21.2 × 3.3 mm
天线 / Antenna	External antenna via connector/module RF path; 50 Ω RF system

重要 / Important: 3.7 V 电池接口是开发板级接口，不代表 ESP32-C5 模组本体可直接接入 3.7 V。模组推荐电源范围为 3.0–3.6 V。 / The 3.7 V battery connector is a board-level interface; it does not mean the ESP32-C5 module itself should be driven directly from 3.7 V. The module recommended supply range is 3.0–3.6 V.

5. 显示与触摸 / 5. Display and Touch

LCD 参数 / LCD Parameters

项目 / Item	规格 / Specification
尺寸 / Size	2.8 inch
驱动 IC / Driver IC	ST7789
原生分辨率 / Native resolution	240 × 320
横屏逻辑分辨率 / Landscape logical resolution	320 × 240 (depending on rotation setting)
颜色顺序 / Color order	BGR
反色 / Inversion	Disabled / OFF; normal display mode
SPI 时钟 / SCK	GPIO6
SPI MOSI	GPIO7
SPI MISO	GPIO2 (shared with TF card; used if required by library/readback)
LCD CS	GPIO10

项目 / Item	规格 / Specification
LCD Backlight / 背光	GPIO13
LCD DC / RS	GPIO14

触摸参数 / Touch Parameters

- 触摸类型：4 线电阻式触摸；控制器：XPT2046。 / Touch type: 4-wire resistive; controller: XPT2046.
- DOUT=GPIO3, SCK=GPIO4, DIN=GPIO5。 / DOUT=GPIO3, SCK=GPIO4, DIN=GPIO5.
- CS=GPIO23, IRQ=GPIO25。 / CS=GPIO23, IRQ=GPIO25.
- 触摸使用独立软件 SPI；检测工具流程标注为无需校准。 / Touch uses an independent software SPI; the supplied test flow is marked as requiring no calibration.

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6. GPIO 映射 / 6. GPIO Mapping

以下映射来自随板提供的客户版 GPIO 电子表格并经原理图核对。板载外设使用 19 个普通 GPIO；TX0=GPIO11、RX0=GPIO12 另用于 UART0；N8R8 的 GPIO15 被 PSRAM 占用，因此 21 个可用 GPIO 已全部分配；EN 不计作 GPIO。

The mapping below is based on the supplied GPIO spreadsheet and cross-checked against the schematic. Board peripherals use 19 GPIOs; TX0=GPIO11 and RX0=GPIO12 are additionally used by UART0; GPIO15 is reserved by PSRAM on N8R8, so all 21 usable GPIOs are allocated; EN is not a GPIO.

GPIO	Signal	定义 / Definition	方向 / Direction	功能 / Function	说明 / Notes
0	GPSTX	GPS 串口 TX / GPS UART TX	Input / 输入	GPS test / GPS 测试	GPS module TX → ESP32-C5
1	BAT_ADC	电池电压检测 / Battery voltage sensing	Input/ADC	Battery test	100k/100k divider; ADC × 2 ≈ battery voltage
2	SD_MISO	TF DAT0 / LCD MISO 共用 / Shared	Input	TF card test	Shared SPI MISO
3	RTP_DOUT	XPT2046 DOUT	Input	Touch test	Touch controller data output
4	RTP_SCK	XPT2046 SCK	Output	Touch test	Independent software SPI
5	RTP_DIN	XPT2046 DIN	Output	Touch test	Independent software SPI
6	LCD_SCK	LCD SCK / TF SCK 共用 / Shared	Output	LCD/TF test	Shared SPI clock
7	LCD_MOSI	LCD MOSI / TF CMD(MOSI) 共用 / Shared	Output	LCD/TF test	Shared SPI MOSI
8	AUDIOPWM	音频 PWM / Audio PWM	Output	Audio test	Audio output control
9	GPSRX	GPS 串口 RX / GPS UART RX	Output	GPS test	ESP32-C5 TX → GPS module RX
10	LCDCS	ST7789 CS	Output	Display test	LCD chip select
13	LCDBL	LCD 背光 / LCD backlight	Output	Display test	Backlight control
14	LCD_RS/DC	ST7789 DC	Output	Display test	Data/command select
23	RTPCS	XPT2046 CS	Output	Touch test	Touch chip select
24	AMP_EN	功放使能 / Amplifier enable	Output	Audio test	High-level enable per supplied mapping; verify actual hardware revision
25	RTPIRQ	XPT2046 IRQ	Input	Touch test	Touch interrupt
26	GREENLED	绿色 LED / Green LED	Output	LED test	HIGH = ON, LOW = OFF
27	SDCS	TF DAT3 / CS	Output	TF card test	TF card SPI chip select

GPIO	Signal	定义 / Definition	方向 / Direction	功能 / Function	说明 / Notes
28	BOOTAUTO	BOOT / 自动启动控制 / auto-start control	Input/Control	Factory restore	BOOT input / factory-restore logic

系统与串口信号 / System and UART Signals

信号 / Signal	用途 / Purpose	备注 / Note
RESET_AUTO / EN	ESP32 reset / enable / ESP32 复位与使能	System enable/reset; not a GPIO / 系统使能/复位，不属于 GPIO
UART0_TX / TX0 / GPIO11	UART0 transmit / UART0 串口发送	Used; excluded from the 19 peripheral GPIO count / 已使用；不计入上述 19 个外设 GPIO
UART0_RX / RX0 / GPIO12	UART0 receive / UART0 串口接收	Used; excluded from the 19 peripheral GPIO count / 已使用；不计入上述 19 个外设 GPIO

7. 接口与总线共用说明 / 7. Interface and Bus Sharing

项目 / Item	GPIO	说明 / Description
LCD + TF SPI 共用 / Shared SPI	GPIO6 / GPIO7 / GPIO2	SCK, MOSI, MISO shared; devices separated by independent CS / 共用时钟与数据线，使用独立 CS 区分设备
LCD CS	GPIO10	LCD_CS
TF card CS	GPIO27	SD_CS / DAT3
XPT2046 touch	GPIO3 / 4 / 5 / 23 / 25	DOUT / SCK / DIN / CS / IRQ
GPS UART	GPIO0 / GPIO9	GPSTX / GPSRX; GPIO0 receives GPS TX, GPIO9 drives GPS RX
Battery ADC	GPIO1	BAT_ADC; 100k/100k divider; C20 100 nF filtering
Green LED	GPIO26	HIGH ON, LOW OFF
Audio	GPIO8 / GPIO24	AUDIOPWM / AMP_EN
LCD control	GPIO13 / GPIO14	Backlight / DC

SPI 共用注意事项 / SPI Sharing Notes

- LCD 与 TF 卡共用 GPIO6(SCK)、GPIO7(MOSI)、GPIO2(MISO)，但分别使用 GPIO10(LCD_CS) 与 GPIO27(SD_CS)。 / LCD and TF card share GPIO6 (SCK), GPIO7 (MOSI), and GPIO2 (MISO), but use GPIO10 (LCD_CS) and GPIO27 (SD_CS) respectively.
- 同一时刻只使能目标设备的 CS，避免多个器件同时选中导致总线冲突。 / Assert only the target device CS at a time to avoid bus contention.
- 触摸 XPT2046 使用独立软件 SPI，不与 LCD/TF 的主 SPI 时钟和 MOSI 共用。 / The XPT2046 touch controller uses an independent software SPI and does not share the main LCD/TF SCK/MOSI signals.

8. 检测功能对应 / 8. Functional Test Mapping

检测项目 / Test	GPIO	检测内容 / What is tested	结果/说明 / Result/Note
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检测项目 / Test	GPIO	检测内容 / What is tested	结果/说明 / Result/Note
屏幕测试 / Display	GPIO6, 7, 10, 13, 14, 2	ST7789 display, 240×320 native / 320×240 landscape; BGR; backlight	Normal / Abnormal
触摸测试 / Touch	GPIO3, 4, 5, 23, 25	XPT2046 touch response	Test flow: no calibration
TF 卡测试 / TF card	GPIO2, 6, 7, 27	SPI init, read/write/format	Shares SPI with LCD
音频测试 / Audio	GPIO8, 24	PWM audio and amplifier enable	Continuous playback depends on firmware
Wi-Fi 测试 / Wi-Fi	Internal RF	2.4 GHz / 5 GHz scan	No listed GPIO occupied
BLE 测试 / BLE	Internal RF	BLE scan	No listed GPIO occupied
GPS 测试 / GPS	GPIO0, 9	GPS UART data	GPSTX / GPSRX
LED 测试 / LED	GPIO26	Green LED	HIGH on / LOW off
电池检测 / Battery	GPIO1	ADC measures divided battery voltage	Voltage estimate, not a coulomb counter
BOOT 恢复出厂 / Factory restore	GPIO28	BOOT input / restore logic	Behavior is defined by firmware

9. 接口与电气注意事项 / 9. Connector and Electrical Notes

电池接口 / Battery Connector

开发板提供 1.25 mm 2-Pin 3.7 V 单节电池接口。按标注图的观察方向，左侧为正极（+），右侧为负极（-）。如需电池供电，建议使用容量 1000–3000 mAh、带保护板（保护电路）的单节 3.7 V 锂离子/锂聚合物电池，并在连接前确认接口规格与正负极。反接可能损坏开发板。

The board provides a 1.25 mm 2-pin connector for a single-cell 3.7 V battery. In the orientation shown in the labeled image, the left pin is positive (+) and the right pin is negative (-). For battery-powered operation, use a protected single-cell 3.7 V Li-ion/LiPo battery; 1000–3000 mAh is recommended. Verify connector type and polarity before connection. Reverse polarity may damage the board.

GPIO1 通过 100 kΩ / 100 kΩ 分压网络检测电池电压，因此在理想情况下，实际电池电压约为 ADC 引脚电压的 2 倍。该功能仅用于电压估算，不是电量库仑计。

GPIO1 senses battery voltage through a 100 kΩ / 100 kΩ divider, so ideally the battery voltage is approximately 2× the ADC pin voltage. This is a voltage estimate only and is not a fuel-gauge/coulomb-counter measurement.

GPS 接口 / GPS Interface

GPS 使用 GPIO0 与 GPIO9 作为 UART 信号。GPIO0（GPSTX 命名）在该板定义中作为输入，连接 GPS 模块 TX；

GPIO9（GPSRX 命名）作为输出，连接 GPS 模块 RX。1.25 mm 4-Pin 连接器的电源脚与实际引脚顺序请以对应硬件版本丝印/原理图为准。

GPS uses GPIO0 and GPIO9 as UART signals. In this board mapping, GPIO0 (named GPSTX) is an input connected to the GPS module TX, while GPIO9 (named GPSRX) is an output connected to the GPS module RX. For the 1.25 mm 4-pin connector power pins and exact physical pin order, follow the silkscreen/schematic of the specific hardware revision.

外置天线 / External Antenna

ESP32-C5-WROOM-1U 为外置天线版本。板上 IPEX1 接口需要连接与 2.4/5 GHz 频段匹配、阻抗为 50 Ω 的天线。请勿在未连接合适天线的情况下长时间进行高功率发射。

ESP32-C5-WROOM-1U is the external-antenna variant. Connect a suitable 2.4/5 GHz, 50-ohm antenna to the IPEX1 connector. Avoid prolonged high-power transmission without a proper antenna attached.

音频接口 / Audio Connector

音频输出由 GPIO8 (AUDIOPWM) 产生 PWM 信号, GPIO24 (AMP_EN) 控制功放使能; 板上顶部为 2.0 mm 2-Pin 音频/扬声器接口。不同负载与音量设置请以实际功放电路能力为准。

Audio output uses GPIO8 (AUDIOPWM) for the PWM signal and GPIO24 (AMP_EN) for amplifier enable. The top connector is a 2.0 mm 2-pin audio/speaker connector. Use loads and volume settings within the capability of the actual amplifier circuit.

10. 软件开发与初始化建议 / 10. Software Development and Initialization Notes

- 显示驱动选择 ST7789, 原生分辨率设为 240×320; 横屏可根据所用图形库设置旋转为 320×240。 / Select ST7789 as the display driver and use 240×320 as the native resolution; set rotation for 320×240 landscape as supported by your graphics library.
- 颜色顺序使用 BGR; 不要启用显示反色 (INVON)。 / Use BGR color order; do not enable display inversion (INVON).
- LCD/TF 共用 SPI 时, 初始化后分别管理 LCD_CS(GPIO10) 与 SD_CS(GPIO27)。 / When LCD and TF share SPI, manage LCD_CS (GPIO10) and SD_CS (GPIO27) independently after initialization.
- XPT2046 使用 GPIO3/4/5/23/25 的独立软件 SPI。 / XPT2046 uses an independent software SPI on GPIO3/4/5/23/25.
- 读取电池电压时, 对 GPIO1 的 ADC 结果按约 2:1 分压比例换算, 并结合 ADC 校准提高精度。 / For battery voltage, convert the GPIO1 ADC reading using the approximately 2:1 divider ratio and apply ADC calibration for better accuracy.
- Wi-Fi 5 GHz 可用信道和发射功率受所在国家/地区法规限制。 / Available 5 GHz Wi-Fi channels and transmit power depend on local regulations.
- BOOT/恢复出厂逻辑由固件决定; 避免在运行中误触发造成配置被清除。 / BOOT/factory-restore behavior is firmware-defined; avoid unintended activation that may clear settings.

建议初始化顺序 / Suggested Initialization Sequence

1. 电源稳定后初始化系统与串口。

After power is stable, initialize the system and debug UART.

2. 初始化 LCD SPI, 设置 ST7789、BGR、关闭反色, 再打开背光。

Initialize LCD SPI; configure ST7789, BGR, inversion off, then enable backlight.

3. 初始化 XPT2046 触摸。

Initialize XPT2046 touch.

4. 如需使用存储功能, 请先插入用户自备的兼容 microSD/TF 卡, 再初始化, 并确保 LCD_CS 与 SD_CS 状态正确。

If removable storage is required, insert a compatible user-supplied microSD/TF card, then initialize it and ensure correct LCD_CS/SD_CS states.

5. 按需求初始化 GPS、音频、LED、电池 ADC。

Initialize GPS, audio, LED, and battery ADC as needed.

6. 最后启用 Wi-Fi/BLE/802.15.4 无线功能。
Enable Wi-Fi/BLE/802.15.4 radio functions as required.

11. 常见问题排查 / 11. Troubleshooting

问题 / Issue	English	处理方法 / Solution	Solution (EN)
屏幕白屏/黑屏	Blank/black display	检查 GPIO6/7/10/13/14、ST7789 驱动选择、BGR 设置、背光 GPIO13 以及反色是否被误开启。	Check GPIO6/7/10/13/14, ST7789 driver selection, BGR setting, backlight GPIO13, and whether inversion was accidentally enabled.
颜色不正确	Incorrect colors	确认图形库颜色顺序为 BGR。	Confirm the graphics library is configured for BGR color order.
触摸无响应	No touch response	检查 GPIO3/4/5/23/25 与 XPT2046 初始化。	Check GPIO3/4/5/23/25 and XPT2046 initialization.
TF 卡初始化失败	TF card init fails	检查 LCD/TF 共用 SPI 的 SCK/MOSI/MISO，以及 SD_CS(GPIO27) 与 LCD_CS(GPIO10) 是否冲突。	Check shared SPI SCK/MOSI/MISO and make sure SD_CS (GPIO27) and LCD_CS (GPIO10) are not asserted simultaneously.
Wi-Fi 信号弱或扫描不到	Weak/no Wi-Fi	确认 IPEX1 外置天线已连接，并使用支持 2.4/5 GHz 的合适天线。	Confirm the IPEX1 external antenna is connected and suitable for 2.4/5 GHz.
电池电压读数约为实际一半	Battery reading is about half	GPIO1 前端为 100k/100k 分压，换算时需要乘以约 2。	GPIO1 uses a 100k/100k divider; multiply the ADC-side voltage by approximately 2.
GPS 无数据	No GPS data	确认 GPIO0 接 GPS TX、GPIO9 接 GPS RX，并核对波特率与供电。	Confirm GPIO0 receives GPS TX, GPIO9 drives GPS RX, and verify baud rate and power.

12. 资料下载与技术支持 / 12. Downloads and Technical Support

ZLX-ESP32-2 的固件、文件、后续资料和相关支持请优先访问 ZLX ESP32 开发板创客支援中心。刷写固件前请确认页面所列 PCB 型号与手中板卡完全一致。

For ZLX-ESP32-2 firmware, files, future documentation, and related support, use the ZLX ESP32 development-board support center. Before flashing firmware, confirm the PCB model shown on the website exactly matches your hardware.

资料下载 / Downloads: <https://www.zlxchina.xyz/>

Espressif module datasheet / 乐鑫模组规格书: [ESP32-C5-WROOM-1 / WROOM-1U Datasheet](#)

参考资料 / References

- ZLX ESP32 开发板创客支援中心 / ZLX ESP32 Development Board Support Center — <https://www.zlxchina.xyz/>
- Espressif ESP32-C5-WROOM-1 & WROOM-1U Datasheet v1.2 (2026-08-12) / Espressif ESP32-C5-WROOM-1 & WROOM-1U Datasheet v1.2 (2026-08-12) — https://espressif.com/documentation/esp32-c5-wroom-1_wroom-1u_datasheet_en.pdf

13. 使用与安全注意事项 / 13. Safety and Handling

- 连接电池前必须确认正负极，禁止反接。 / Verify battery polarity before connection; never reverse the battery connector.
- 插拔 IPEX 天线接头时保持垂直受力，避免侧向撬动造成射频座脱落。 / When mating/unmating the IPEX connector, apply force vertically and avoid sideways prying that may damage the RF socket.
- 开发板裸板使用时注意静电防护，避免金属物造成短路。 / Use ESD precautions with the bare board and prevent metal objects from causing shorts.
- 5 GHz 无线频段、信道和功率必须符合使用地区的法规。 / 5 GHz radio band, channels, and transmit power must comply with local regulations.
- TF 卡、GPS、音频等外设的供电能力与电平请遵循实际硬件版本，未确认前不要向 IO 引脚施加高于 3.3 V 的电平。 / For TF, GPS, audio, and other peripherals, follow the actual hardware revision for supply and logic levels. Do not apply logic levels above 3.3 V to IO pins unless explicitly verified.

14. 文档版本记录 / 14. Document Revision History

版本 / Rev.	日期 / Date	内容 / Description	来源 / Basis
V1.1	2026-08-25	修订：microSD/TF、电池与 GPIO 说明。 Revised microSD/TF, battery, and GPIO notes.	Hefei Zhiluxing Information Technology Co., Ltd.

— End / 结束 —